

Open Sidebar

<https://api.electricityinfo.co.nz/api/quantities/v1> - Live

[Authorize](#)



Energy and Reserve Quantities

0.0.1

 [View Raw](#)

OAS3

PRS schedule, NI and SI (rolling window -24 to +24 TP)

Pre-requisites

See [How To Call Our APIs](#) for information on registration, authentication and authorisation requirements.

Overview

The purpose of this API is to retrieve energy and reserve quantities for identified schedule(s).

Quantity information can be filtered in a variety of different ways. The minimum required parameters to filter data are a list of one or more schedules (market run types) and the market type being queried (Energy or Reserves).

- Energy quantities for a single schedule can be queried through the `/schedules/{schedule}/energy` path.
- Reserve quantities for a single schedule can be queried through the `/schedules/{schedule}/reserves` path.
- Energy quantities for multiple schedules can be queried through the `/energy` path, identifying one or more schedules in the `schedules` query parameter.
- Reserve quantities for multiple schedules can be queried through the `/reserves` path, identifying one or more schedules in the `schedules` query parameter. Schedules supported by this API can be obtained through a query to `/schedules`.
- Check out the WITS portal: <https://www2.electricityinfo.co.nz/>

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Reserve Quantities must include a **runClass** filter. This query parameter must be set to one of **InstantaneousReserve** , **ReserveOffers** or **AdjustedReserveOffers** .

Range Requests

Quantity data can be filtered using date-times for absolute windows, or by period offsets for rolling windows.

The available query parameters for specifying a range are:

- **from**
- **to**
- **back**
- **forward**

From and To

The **from** and **to** parameters specify a date-time for which to filter queried data by. None, one or both parameters may be provided. The date-time must conform to the **RFC3339** standard formatting, e.g. **yyyy-MM-dd'T'HH:mm:ssXXX** .

If both **from** and **to** are provided, this is an inclusive window of time to query data for.

If a **from** parameter is provided without a corresponding **to** parameter, this represents a query of data filtered from the trading period implied by the **from** date-time forward as far as possible for the queried schedule(s).

If a **to** parameter is provided without a corresponding **from** parameter, this represents a query of data filtered from the oldest data available forward to the trading period implied by the **to** date-time.

Back and Forward

Unlike the **from** and **to** parameters, the **back** and **forward** filter can be used to identify a sliding window of data to query. A request cannot include both **from** and/or **to** parameters **and** **back** and/or **forward** .

The **back** parameter specifies a number of trading periods *before* the current trading period to include data for. Correspondingly, the **to** parameter specifies a number of trading periods *ahead* of the current trading period to include data for.

For example, if the current trading period is **23** (11:00:00 NZT - 11:29:59 NZT) and both **back** and **forward** are set to **5** , data from trading periods **18** - **28** will be queried. Note that not all schedules will have data available beyond the current trading period.



Menu

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Energy



Retrieve a list of schedules for which quantity data is currently available



GET /schedules

Parameters

Try it out

No parameters

X

Use 'Try it Out' to see completed code snippet

Responses

Successfully listed schedules

No links

Media type

application/json

Controls Accept header.

[Example Value](#) [Schema](#)

200

```
{
  "reserveSchedules": [
    "NRS",
    "PRS"
  ],
  "energySchedules": [
    "NRS",
    "PRS",
```

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Client has supplied invalid parameters

No links

Media type

Examples

application/json

Invalid API Request

[Example Value](#) [Schema](#)

400

```
{
  "status": 400,
  "code": "BAD_REQUEST",
  "message": "Invalid Request",
  "detail": "52 is greater than maximum 50 for paramter
\"tradingPeriod\"",
  "timestamp": "20210406T11:54:33+13:00"
}
```

Resource not found

No links

Media type

Examples

application/json

Client not authorised to access resource

[Example Value](#) [Schema](#)

403

```
{
  "status": 403,
  "code": "FORBIDDEN",
  "message": "Invalid or missing authorisation token",
  "timestamp": "20210406T11:54:33+13:00"
}
```

Method not allowed

No links

Media type

Examples

application/json

Invalid accept header

[Example Value](#) [Schema](#)

405

```
{
  "status": 405,
  "code": "METHOD_NOT_ALLOWED",
}
```

Open Sidebar

Content type not produced

NO LINKS

Media type

Examples

application/json

Invalid accept header

[Example Value](#) [Schema](#)

406

```
{
  "status": 406,
  "code": "NOT_ACCEPTABLE",
  "message": "Requested content type \"application/xml\" is not produced
by this API",
  "detail": "Only \"application/json\" is produced by this API",
  "timestamp": "20210406T11:54:33+13:00"
}
```

Content type not produced

No links

Media type

Examples

application/json

Invalid accept header

[Example Value](#) [Schema](#)

500

```
{
  "status": 500,
  "code": "INTERNAL_SERVER_ERROR",
  "message": "Internal server error. Please try again later.",
  "timestamp": "20210406T11:54:33+13:00"
}
```

Retrieve a list of energy quantities for the given schedule



GET **/schedules/{schedule}/energy**

Parameters

Try it out



Open Sidebar

from	Optional trading date-time to filter historical data
string	
(query)	from - Optional trading date-time to filter historical data
to	Optional trading date-time to filter historical data
string	
(query)	to - Optional trading date-time to filter historical data
back	Optional trading period filter
integer	
(query)	back - Optional trading period filter
forward	Optional trading period filter
integer	
(query)	forward - Optional trading period filter
island	Return quantity data only for the selected island
string	
(query)	Available values : NI, SI --

X

Use 'Try it Out' to see completed code snippet

Responses

Successfully received energy quantities

No links

Media type

application/json

Controls Accept header.

Open Sidebar

```

    "tradingPeriod": 2,
    "schedule": "NRS",
    "island": "NI",
    "load": 4191.901,
    "generation": 2829.59,
    "intermittentGeneration": 813.578,
    "totalBids": 2324,
    "totalOffers": 8465,
    "intermittentOffers": 165
  }
]
}

```

Client has supplied invalid parameters

No links

Media type

Examples

application/json

Invalid API Request

[Example Value](#) [Schema](#)

400

```

{
  "status": 400,
  "code": "BAD_REQUEST",
  "message": "Invalid Request",
  "detail": "52 is greater than maximum 50 for paramter
\"tradingPeriod\"",
  "timestamp": "20210406T11:54:33+13:00"
}

```

Resource not found

No links

Media type

Examples

application/json

Client not authorised to access resource

403

[Example Value](#) [Schema](#)

```

{
  "status": 403,
  "code": "FORBIDDEN",
  "message": "Invalid or missing authorisation token",
}

```

Open Sidebar

Media type	Examples
application/json	Schedule not found

404 [Example Value](#) [Schema](#)

```
{
  "status": 404,
  "code": "NOT_FOUND",
  "message": "Schedule \"ABC\" not found",
  "timestamp": "20210406T11:54:33+13:00"
}
```

Method not allowed

No links

Media type	Examples
application/json	Invalid accept header

405 [Example Value](#) [Schema](#)

```
{
  "status": 405,
  "code": "METHOD_NOT_ALLOWED",
  "message": "HTTP Method \"PUT\" is not valid for this operation",
  "timestamp": "20210406T11:54:33+13:00"
}
```

Content type not produced

No links

Media type	Examples
application/json	Invalid accept header

406 [Example Value](#) [Schema](#)

```
{
  "status": 406,
  "code": "NOT_ACCEPTABLE",
  "message": "Requested content type \"application/xml\" is not produced by this API",
  "detail": "Only \"application/json\" is produced by this API",
}
```



Open Sidebar

Media type

application/json

Examples

Invalid accept header

500

[Example Value](#) [Schema](#)

```
{
  "status": 500,
  "code": "INTERNAL_SERVER_ERROR",
  "message": "Internal server error. Please try again later.",
  "timestamp": "20210406T11:54:33+13:00"
}
```

Retrieve a list of energy quantities across schedules



GET /energy

Parameters

Try it out

schedules

Required

array[string]

(query)

from

string

(query)

Optional trading date-time to filter historical data

from - Optional trading date-time to filter historical data

to

string

(query)

Optional trading date-time to filter historical data

to - Optional trading date-time to filter historical data

back

integer

Optional trading period filter



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forward - Optional trading period filter

forward - Optional trading period filter

island

string

Return quantity data only for the selected island

(query)

Available values : NI, SI

--

X

Use 'Try it Out' to see completed code snippet

Responses

Successfully received energy quantities

No links

Media type

application/json

Controls Accept header.

[Example Value](#) [Schema](#)

200

```
{
  "schedules": [
    {
      "schedule": "NRS",
      "energyQuantities": [
        {
          "tradingDateTime": "2021-06-16T10:30:00+12:00",
          "tradingPeriod": 2,
          "schedule": "NRS",
          "island": "NI",
          "load": 4191.901,
          "generation": 2829.59,
          "intermittentGeneration": 813.578,
          "totalBids": 2324,
          "totalOffers": 8465,
          "intermittentOffers": 165
        }
      ]
    }
  ]
}
```

Open Sidebar

Media type

Examples

application/json

Invalid API Request

[Example Value](#)
[Schema](#)

400

```
{
  "status": 400,
  "code": "BAD_REQUEST",
  "message": "Invalid Request",
  "detail": "52 is greater than maximum 50 for paramter
  \"tradingPeriod\"",
  "timestamp": "20210406T11:54:33+13:00"
}
```

Resource not found

No links

Media type

Examples

application/json

Client not authorised to access resource

[Example Value](#)
[Schema](#)

403

```
{
  "status": 403,
  "code": "FORBIDDEN",
  "message": "Invalid or missing authorisation token",
  "timestamp": "20210406T11:54:33+13:00"
}
```

Method not allowed

No links

Media type

Examples

application/json

Invalid accept header

[Example Value](#)
[Schema](#)

405

```
{
  "status": 405,
  "code": "METHOD_NOT_ALLOWED",
  "message": "HTTP Method \"PUT\" is not valid for this operation",
}
```

Open Sidebar

Media type

Examples

application/json

Invalid accept header

[Example Value](#)
[Schema](#)

406

```
{
  "status": 406,
  "code": "NOT_ACCEPTABLE",
  "message": "Requested content type \"application/xml\" is not produced
by this API",
  "detail": "Only \"application/json\" is produced by this API",
  "timestamp": "20210406T11:54:33+13:00"
}
```

Content type not produced

No links

Media type

Examples

application/json

Invalid accept header

[Example Value](#)
[Schema](#)

500

```
{
  "status": 500,
  "code": "INTERNAL_SERVER_ERROR",
  "message": "Internal server error. Please try again later.",
  "timestamp": "20210406T11:54:33+13:00"
}
```

Reserves



Retrieve a list of schedules for which quantity data is currently available



GET /schedules



Open Sidebar

X

Use 'Try it Out' to see completed code snippet

Responses

Successfully listed schedules

No links

Media type

application/json

Controls Accept header.

[Example Value](#) [Schema](#)

200

```
{
  "reserveSchedules": [
    "NRS",
    "PRS"
  ],
  "energySchedules": [
    "NRS",
    "PRS",
    "WDS"
  ]
}
```

Client has supplied invalid parameters

No links

Media type

application/json

Examples

Invalid API Request

[Example Value](#) [Schema](#)

400

```
{
  "status": 400,
  "code": "BAD_REQUEST",
  "message": "Invalid Request",
  "detail": "52 is greater than maximum 50 for paramter\n\"tradingPeriod\"",
}
```

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Media type	Examples
application/json	Client not authorised to access resource

403 [Example Value](#) [Schema](#)

```
{
  "status": 403,
  "code": "FORBIDDEN",
  "message": "Invalid or missing authorisation token",
  "timestamp": "20210406T11:54:33+13:00"
}
```

Method not allowed

No links

Media type	Examples
application/json	Invalid accept header

405 [Example Value](#) [Schema](#)

```
{
  "status": 405,
  "code": "METHOD_NOT_ALLOWED",
  "message": "HTTP Method \"PUT\" is not valid for this operation",
  "timestamp": "20210406T11:54:33+13:00"
}
```

Content type not produced

No links

Media type	Examples
application/json	Invalid accept header

406 [Example Value](#) [Schema](#)

```
{
  "status": 406,
  "code": "NOT_ACCEPTABLE",
  "message": "Requested content type \"application/xml\" is not produced by this API",
  "detail": "Only \"application/json\" is produced by this API",
}
```



Open Sidebar

Media type

application/json

Examples

Invalid accept header

500

[Example Value](#) [Schema](#)

```
{
  "status": 500,
  "code": "INTERNAL_SERVER_ERROR",
  "message": "Internal server error. Please try again later.",
  "timestamp": "20210406T11:54:33+13:00"
}
```

Retrieve a list of reserve quantities for the given schedule



GET /schedules/{schedule}/reserves

Parameters

Try it out

schedule

Required

NRS

string

(path)

runClass

Required

Mandatory run class filter

string

(query)

Available values : InstantaneousReserve, ReserveOffers, AdjustedReserveOffers

InstantaneousReserve

from

string

(query)

Optional trading date-time to filter historical data

from - Optional trading date-time to filter historical data

to

string

Optional trading date-time to filter historical data



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back - Optional trading period filter	
forward integer (query)	Optional trading period filter forward - Optional trading period filter
island string (query)	Return quantity data only for the selected island Available values : NI, SI --

X

Use 'Try it Out' to see completed code snippet

Responses

		No links
	Successfully received reserve quantities	
	Media type application/json Controls Accept header. Example Value Schema <pre>{ "schedule": "NRS", "reserveQuantities": [{ "tradingDateTime": "2021-06-16T10:30:00+12:00", "tradingPeriod": 2, "schedule": "NRS", "island": "NI", "runType": "N", "reserveClass": "F", "runClass": "R2", "price": 12.34, "reserveMw": 344.773, </pre>	
200		

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Client has supplied invalid parameters

[No links](#)

Media type

Examples

application/json

Invalid API Request

[Example Value](#) [Schema](#)

400

```
{
  "status": 400,
  "code": "BAD_REQUEST",
  "message": "Invalid Request",
  "detail": "52 is greater than maximum 50 for paramter
\"tradingPeriod\"",
  "timestamp": "20210406T11:54:33+13:00"
}
```

Resource not found

[No links](#)

Media type

Examples

application/json

Client not authorised to access resource

403

[Example Value](#) [Schema](#)

```
{
  "status": 403,
  "code": "FORBIDDEN",
  "message": "Invalid or missing authorisation token",
  "timestamp": "20210406T11:54:33+13:00"
}
```

Resource not found

[No links](#)

Media type

Examples

application/json

Schedule not found

404

[Example Value](#) [Schema](#)

Open Sidebar

Method not allowed

No links

Media type

Examples

application/json

Invalid accept header

405

[Example Value](#) [Schema](#)

```
{
  "status": 405,
  "code": "METHOD_NOT_ALLOWED",
  "message": "HTTP Method \"PUT\" is not valid for this operation",
  "timestamp": "20210406T11:54:33+13:00"
}
```

Content type not produced

No links

Media type

Examples

application/json

Invalid accept header

406

[Example Value](#) [Schema](#)

```
{
  "status": 406,
  "code": "NOT_ACCEPTABLE",
  "message": "Requested content type \"application/xml\" is not produced by this API",
  "detail": "Only \"application/json\" is produced by this API",
  "timestamp": "20210406T11:54:33+13:00"
}
```

Content type not produced

No links

Media type

Examples

application/json

Invalid accept header

500

[Example Value](#) [Schema](#)



Open Sidebar

Retrieve a list of reserve quantities across schedules



GET /reserves

Parameters

Try it out

schedules

Required

array[string]

(query)

runClass

Required

Mandatory run class filter

string

Available values : InstantaneousReserve, ReserveOffers, AdjustedReserveOffers

(query)

InstantaneousReserve

from

string

Optional trading date-time to filter historical data

(query)

from - Optional trading date-time to filter historical data

to

string

Optional trading date-time to filter historical data

(query)

to - Optional trading date-time to filter historical data

back

integer

Optional trading period filter

(query)

back - Optional trading period filter

forward

integer

Optional trading period filter

(query)

Open Sidebar

--

X

Use 'Try it Out' to see completed code snippet

Responses

No links

Successfully received reserve quantities

Media type

application/json

Controls Accept header.

[Example Value](#)
[Schema](#)

200

```

{
  "schedules": [
    {
      "schedule": "NRS",
      "reserveQuantities": [
        {
          "tradingDateTime": "2021-06-16T10:30:00+12:00",
          "tradingPeriod": 2,
          "schedule": "NRS",
          "island": "NI",
          "runType": "N",
          "reserveClass": "F",
          "runClass": "R2",
          "price": 12.34,
          "reserveMw": 344.773,
          "riskMw": 12.43,
          "riskAdjustmentFactor": 33.43
        }
      ]
    }
  ]
}

```

Open Sidebar

[Example Value](#) [Schema](#)

400

```
{
  "status": 400,
  "code": "BAD_REQUEST",
  "message": "Invalid Request",
  "detail": "52 is greater than maximum 50 for paramter
\"tradingPeriod\"",
  "timestamp": "20210406T11:54:33+13:00"
}
```

Resource not found

No links

Media type

Examples

application/json

Client not authorised to access resource

403

[Example Value](#) [Schema](#)

```
{
  "status": 403,
  "code": "FORBIDDEN",
  "message": "Invalid or missing authorisation token",
  "timestamp": "20210406T11:54:33+13:00"
}
```

Method not allowed

No links

Media type

Examples

application/json

Invalid accept header

405

[Example Value](#) [Schema](#)

```
{
  "status": 405,
  "code": "METHOD_NOT_ALLOWED",
  "message": "HTTP Method \"PUT\" is not valid for this operation",
  "timestamp": "20210406T11:54:33+13:00"
}
```

Open Sidebar

[Example Value](#) [Schema](#)

406

```
{
  "status": 406,
  "code": "NOT_ACCEPTABLE",
  "message": "Requested content type \"application/xml\" is not produced
by this API",
  "detail": "Only \"application/json\" is produced by this API",
  "timestamp": "20210406T11:54:33+13:00"
}
```

Content type not produced

No links

Media type

Examples

application/json

Invalid accept header

500

[Example Value](#) [Schema](#)

```
{
  "status": 500,
  "code": "INTERNAL_SERVER_ERROR",
  "message": "Internal server error. Please try again later.",
  "timestamp": "20210406T11:54:33+13:00"
}
```

Schemas



```
listSchedulesResponse {
  reserveSchedules* [...]
  energySchedules* [...]
}
```



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```
getReserveQuantitiesResponse  {  
  schedules*                  [...]  
}
```

```
energyScheduleDetails  {  
  schedule*                schedule string  
                           example: NRS  
  energyQuantities*        [...]  
}
```

```
reserveScheduleDetails  {  
  schedule*                schedule string  
                           example: NRS  
  reserveQuantities        [...]  
}
```

Open Sidebar

maximum: 50

example: 2

A sequential 30-minute period starting from period 1 at midnight (00:00) and, on most days, ending at period 48 at 23:30. The exceptions to this are on the spring and autumn daylight-time changeover days. The spring change-over day is 23 hours long (as the clock jumps forward from 2am to 3am) so only has 46 trading-periods and the autumn change-over day is 25 hours long (as the clock jumps back from 3am to 2am) and has 50 trading-periods.

schedule*

schedule string

example: NRS

island

island string

Enum:

Array [2]

load

number

example: 4191.901

generation

number

example: 2829.59

intermittentGeneration number

example: 813.578

totalBids

number

example: 2324

totalOffers

number

example: 8465

intermittentOffers

number

example: 165

}

Open Sidebar

maximum: 50

example: 2

A sequential 30-minute period starting from period 1 at midnight (00:00) and, on most days, ending at period 48 at 23:30. The exceptions to this are on the spring and autumn daylight-time changeover days. The spring change-over day is 23 hours long (as the clock jumps forward from 2am to 3am) so only has 46 trading-periods and the autumn change-over day is 25 hours long (as the clock jumps back from 3am to 2am) and has 50 trading-periods.

```

schedule*      schedule string
                  example: NRS

island        island string
                  Enum:

                      Array [ 2 ]

runType       string
                  maxLength: 1
                  example: N

reserveClass string
                  example: F

runClass     string
                  example: R2

price        price number
                  multipleOf: 0.01
                  minimum: -1000000000
                  maximum: 1000000000
                  exclusiveMinimum: true
                  exclusiveMaximum: true
                  example: 12.34

reserveMw    number
                  example: 344.773

riskMw       number
                  example: 12.43

riskAdjustmentFactor number
                  example: 33.43
    }
  
```

Open Sidebar

maximum: 50

example: 2

A sequential 30-minute period starting from period 1 at midnight (00:00) and, on most days, ending at period 48 at 23:30. The exceptions to this are on the spring and autumn daylight-time changeover days. The spring change-over day is 23 hours long (as the clock jumps forward from 2am to 3am) so only has 46 trading-periods and the autumn change-over day is 25 hours long (as the clock jumps back from 3am to 2am) and has 50 trading-periods.

schedule*

schedule string

example: NRS

island

island string

Enum:

Array [2]

}

schedule string

example: NRS

island string

Enum:

Array [2]

runClass string

Enum:

Array [3]

tradingDateTime string(\$datetime)

example: 2021-06-16T10:30:00+12:00

Open Sidebar

A sequential 30-minute period starting from period 1 at midnight (00:00) and, on most days, ending at period 48 at 23:30. The exceptions to this are on the spring and autumn daylight-time changeover days. The spring change-over day is 23 hours long (as the clock jumps forward from 2am to 3am) so only has 46 trading-periods and the autumn change-over day is 25 hours long (as the clock jumps back from 3am to 2am) and has 50 trading-periods.

price number

multipleOf: 0.01

minimum: -1000000000

maximum: 1000000000

exclusiveMinimum: true

exclusiveMaximum: true

example: 12.34

fault {

description:

Standard fault model

status*

integer

minimum: 100

maximum: 599

example: 404

The HTTP Status code of the response

code*

string

example: INVALID_NODE

Application specific error code

message*

string

example: Schedule 'XYZ' is not valid

Basic error message

detail

string

Extended error details

timestamp*

string(\$datetime)

example: 20210325T10:00:00+13:00

Server timestamp of failure

}



Menu

Open Sidebar